

SECTION SYLLABUS
ARE 212: MULTIPLE EQUATION ESTIMATION
Spring 2016

GSI Info

GSI: Fiona Burlig

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Course Info

PROFESSOR: Max Auffhammer

LECTURES: T/Th 10A-12P (141 Giannini)

WEBSITE: <http://bcourses.berkeley.edu>

The main goal of this section is to introduce you to R as a tool for econometric analysis. Every section, we'll code up an example of the use of R with real data and some concepts that Max covers in class. The whole point of section is to help you learn to code, so I strongly encourage you to bring your laptops to section and to interrupt me with questions and comments.

Before we get any further, we all owe a great deal of thanks to Dan Hammer, Patrick Baylis, and Kenny Bell, who wrote notes for the 2013, 2014, and 2015 versions of this course, respectively.¹ My notes will contribute to an ongoing repository of ARE 212 notes, so please point out mistakes and make suggestions.

I plan on loosely following the outline of topics below. The actual progression of sections will likely change, but I'll post notes for each section on bCourses by the preceding Monday evening. I'll also post work-in-progress notes on my webpage.²

¹These notes can be found at <https://github.com/danhammer/ARE212>, <https://github.com/pbaylis/ARE212>, and <https://github.com/kendonb/ARE212>. Dan's notes are largely a subset of Patrick's; Kenny's contain some similar material to the other two, but also emphasize some productivity tools that I won't cover. I highly recommend all three sets.

²<http://www.fionaburlig.com/teaching/are212>, as well as useful things I run across. Not like this is a ploy to get more website traffic, or anything.

January 21	Introduction to R
January 28	knitr and matrix operations
February 4	Functions and looping (feat. OLS)
February 11	Goodness of fit
February 18	Hypothesis testing
February 25	Figures with ggplot2
March 3	GLS
March 10	Maximum Likelihood Estimation
March 17	Large-sample properties of OLS
March 24	<i>Spring Break</i>
March 31	Standard errors I
April 7	Standard errors II
April 14	Instrumental variables
April 21	Web scraping
April 28	Spatial data analysis

Email policy: Feel free to email me questions about the course, section, problem sets, your research, or life in general. If I can't answer your question in less than a paragraph, I'll ask you to come to office hours or to make an appointment with me so that we can discuss it in person. If you can't do that, you'll have to come up with a shorter question.³ I will do my best to reply to emails within 48 hours (M-F). If you have not heard from me within 48 hours, please resend the email - it's possible the gnomes living in my inbox ate it. To greatly increase the likelihood that I don't lose your email in my inbox, please include "[ARE 212]" in the title. I will not respond to emails sent within 24 hours of a problem set or exam deadline.⁴

Feedback: I will find your candid feedback on the content, presentation, and pace of section immensely valuable. Please feel free to let me know how things are going in person, via email, or anonymously. I've set up an anonymous feedback form here: <http://goo.gl/forms/qUjDdYBstL>; if you prefer, you may also leave me a note in my Giannini mailbox.

Laptops: As mentioned above, I strongly encourage you to bring your laptop to section with you to follow along with the examples I'm going over. We all know that having a wifi-enabled laptop can be dangerous. I'll do my best to be more interesting than Facebook. If you find yourself having too many twitter followers, feel free to live-tweet section.⁵

Problem sets: See Max's syllabus for details on problem sets and grading. I'll just add a not-so-gentle plea for you to carefully comment your code. In the short run, commenting your code will make you think carefully about what you're doing as you're doing it. In the medium run, well-commented code will make it much easier for your future self to look back over your code and understand what you did (which, rumor has it, helps on the midterm/final). In the long run, taking the time to comment your code will make you a better econometrician and collaborator throughout your career. Finally, commenting your code will make my grading life significantly easier, improving my mood while I'm grading your problem sets and allowing

³Read: for questions about your research or life in general, we should talk in person. I don't bite.

⁴My wife is a early-to-bed early-to-rise type, and therefore I am too. I put this clause in here to save you the pain of sending me a frantic email the night before something is due and expecting, but not receiving, a response.

⁵Please, for the sake of your own internet street cred, do not livetweet section.

me to give you more useful feedback.

Attendance: Attendance at section is *not* mandatory. I hope that section will be helpful, but I don't expect that they'll be useful for everyone. Show up if $MB_{\text{section}} > MC_{\text{section}}$ in any given week.

Additional resources: There are a variety of resources that teach R, econometrics, or both. Here are a couple of my favorites (I'll post more on my ARE 212 website as the semester progresses):

1. Google's R style guide: <https://google.github.io/styleguide/Rguide.xml>
2. Swirl (R tutorials): <http://swirlstats.com/>
3. R reference card: <http://cran.r-project.org/doc/contrib/refcard.pdf>
4. R search engine: <http://www.rseek.com>
5. R's built-in help function: `?()`
6. Stack Overflow: <http://stackoverflow.com/questions/tagged/r>

Special accommodations: If you require any disability-related accommodations, please see me privately as soon as possible. You will ultimately need to acquire an accommodations letter from the Disabled Students Program (<http://dsp.berkeley.edu>), which will be sent directly to Max.

Academic integrity: This should go without saying, but this class will be conducted in strict observance of the University's Academic Honesty Code (<http://asuc.org/honorcode/>) and the University's Code of Student Conduct (<http://sa.berkeley.edu/code-of-conduct>), without compromise or exception.